The unit circle

1

a Quadrant 4 **b** Quadrant 1 **c** Quadrant 2 **d** Quadrant 3

e Quadrant 1 **f** Quadrant 4 **g** Quadrant 3 **h** Quadrant 2

2

a

i Quadrants 1 and 2 **ii** Quadrants 3 and 4

b

i Quadrants 1 and 4 **ii** Quadrants 2 and 3

c

i Quadrants 1 and 3 **ii** Quadrants 2 and 4

3

a Negative **b** Positive **c** Positive **d** Negative

e Positive **f** Positive **g** Negative **h** Negative

i Negative

4

a False **b** False **c** True **d** True

e True

5

a Quadrant 1 **b** Quadrant 4

6

a Quadrant 2 **b** Quadrant 2 **c** Quadrant 2 **d** Quadrant 1

Equivalent angles

7 30°

8

a

i $Q(-a, b)$ **ii** $R(-a, -b)$ **iii** $S(a, -b)$

b

i $\sin 49^\circ$ **ii** $-\sin 49^\circ$ **iii** $-\sin 49^\circ$

c

i $\sin x$ **ii** $-\sin x$ **iii** $-\sin x$

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9

a

i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b

i $-\cos 63^\circ$ ii $-\cos 63^\circ$ iii $\cos 63^\circ$

c

i $-\cos x$ ii $-\cos x$ iii $\cos x$

10

a

i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b

i $-\tan 62^\circ$ ii $\tan 62^\circ$ iii $-\tan 62^\circ$

c

i $-\tan x$ ii $\tan x$ iii $-\tan x$

11

a

i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$

b

i $\frac{20}{29}$ ii $-\frac{21}{29}$ iii $-\frac{20}{21}$

c

i $-\frac{21}{29}$ ii $\frac{20}{29}$ iii $-\frac{21}{20}$

12

a 65° **b** 30° **c** 60° **d** 75°

13

a $\sin 87^\circ$ **b** $-\cos 15^\circ$ **c** $-\tan 61^\circ$ **d** $-\sin 41^\circ$

e $-\cos 42^\circ$ **f** $\tan 71^\circ$

14

a $\sin\theta$ **b** $\cos\theta$ **c** $\sin\theta$ **d** $-\cos\theta$

e $-\sin\theta$ **f** $\cos\theta$ **g** $-\sin\theta$ **h** $\cos\theta$

15

a 0.93 **b** 0.36 **c** -0.93 **d** 0.36

e -0.36 **f** -0.93 **g** 0.36 **h** -0.93

16

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a 0.56 **b** 0.49 **c** 0.89

17

a $P(0.41, 0.91)$ **b** $(-0.41, 0.91)$ **c** 0.91 **d** -0.41

e -2.25

18

a $\sin\theta = -\frac{4}{5}$ **b** $\tan\theta = -\frac{4}{3}$

19

a $\cos\theta = \pm\frac{3}{4}$ **b** $\tan\theta = \pm\frac{\sqrt{7}}{3}$

20

a $\sin\theta = \frac{3}{5}$ **b** $\cos\theta = -\frac{4}{5}$ **c** $\tan\theta = -\frac{3}{4}$

21

a $\cos\theta = \frac{10}{\sqrt{269}}$ **b** $\sin\theta = -\frac{13}{\sqrt{269}}$

22 $\cos\theta = -\frac{8}{17}$

23

a $\sin\theta = \frac{11}{61}$ **b** $\sin\theta = \frac{\sqrt{13}}{7}$

24

a $\tan\theta = \frac{2\sqrt{10}}{3}$ **b** $\tan\theta = \frac{1}{3}$

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Exact trigonometric ratios

25

a

i $\frac{\sqrt{3}}{2}$ ii $\frac{1}{2}$ iii $\sqrt{3}$

b

i 120° ii 240° iii 300°

c

i $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ii $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ iii $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

26

a

i $\frac{1}{2}$ ii $\frac{\sqrt{3}}{2}$ iii $\frac{1}{\sqrt{3}}$

b

i 150° ii 210° iii 330°

c

i $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ii $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ iii $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

27

a 30° **b** $-\frac{\sqrt{3}}{2}$

28

a $-\frac{1}{\sqrt{2}}$ **b** $-\frac{1}{\sqrt{2}}$ **c** 1 **d** $-\frac{1}{2}$

e $\frac{1}{\sqrt{2}}$ **f** $\frac{1}{\sqrt{2}}$ **g** $-\frac{1}{\sqrt{3}}$ **h** $-\frac{\sqrt{3}}{2}$

i $\frac{\sqrt{3}}{2}$ **j** $-\frac{1}{\sqrt{3}}$ **k** 0 **l** $\sqrt{3}-1$

m $\frac{15}{4}$

29 $-\frac{1}{4}$

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